

Spinocerebellar ataxia 3 (SCA3) is a neurodegenerative disease that manifests as a progressive loss of coordination of the limbs, and frequently results in impaired speech and eye coordination. It is associated with extension of the polyglutamine (polyQ) region of the ataxin-3 protein. The polyQ region is encoded by a series of consecutive CAG codons known as poly-CAG repeats. These regions vary in length from individual to individual, ranging from as few as 10 to as many as 80 repeats. SCA3 has been reported in patients with polyQ regions exceeding ~55 repeats, and increased polyQ length correlates with earlier onset of the disease and greater clinical severity.

Circular dichroism (CD) is a method of assessing protein secondary structure. It measures a molecule's ability to absorb left- and right-handed circularly polarized light of varying wavelengths. The absorption difference is known as "ellipticity," denoted by $\Delta\epsilon$. Alpha-helices have a maximum $\Delta\epsilon$ at 190 nm, and beta-sheets have a maximum $\Delta\epsilon$ at 200 nm.

CD results show that as the length of the ataxin-3 polyQ region increases, alpha-helical nature is lost and the protein aggregates, ultimately precipitating out of solution and depositing in brain and other tissues. Furthermore, as proteins aggregate, they form beta-sheets. These beta-sheets are in the parallel orientation, and each sheet can interact with sheets in other proteins to form long, insoluble structures known as amyloid fibers (Figure 1).

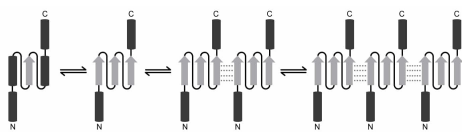


Figure 1 Depiction of the ataxin-3 polyQ region shifting from alpha-helical (cylinders) to beta-sheet (arrows) structure and aggregating. Dashed lines represent interactions between hydrophobic residues of adjacent ataxin-3 proteins.

Glutamine can be encoded by both CAG and CAA codons. Researchers investigated survival rates in fruit flies expressing ataxin-3 with polyQ regions of equal length encoded by either CAG-CAG or CAA-CAG repeats (Table 1). The protein products had identical amino acid sequences.

Table 1 Fruit Fly Survival Rates Relative to Ataxin-3 Expression

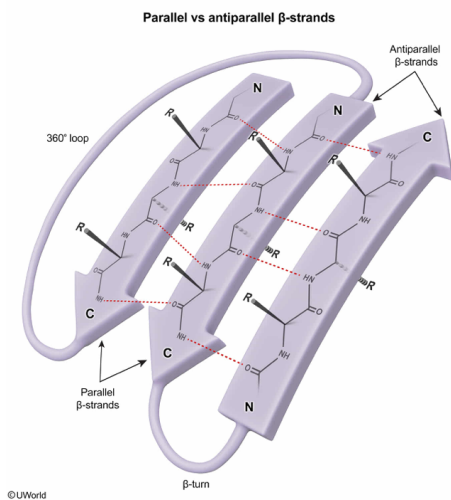
Poly-Q region encoded by	Relative ataxin-3 levels	% ataxin-3 aggregated	% flies alive after 20 days
Control (normal polyQ region)	1	0%	100%
CAA-CAG repeats	1.3	33%	84%
CAG-CAG repeats	1.1	32%	5%

Which statements regarding ataxin-3 parallel beta-sheets are true?

- I. Short sequences called beta-turns link adjacent beta-strands to each other.
- II. The N-terminal portion of each beta-strand is aligned with the N-terminal portions of adjacent strands.
- III. The backbone carbonyl and NH groups of adjacent beta-strands form hydrogen bonds with each other.
- ☐ A. I only [3%]
- ☐ B. I and II only [30%]
- ☒ C. II and III only [36%]
- ☐ D. I and III only [29%]

Correct
37% answered correctly

Explanation:



Beta-sheets can be oriented in either a **parallel** or an **antiparallel** manner. The passage states that the sheets in ataxin-3 are in the parallel orientation. **Parallel strands run in the same direction**, so the N-terminal portion of one strand aligns with the N-terminal portions of the others (**Number II**). On the other hand, in antiparallel sheets the individual strands run in directions opposite to each other, so the N-terminal portion of one strand lines up with the C-terminal portion of neighboring strands.

Secondary structure always includes **hydrogen bonds between amide carbonyls and NH groups** in the polypeptide backbone (**Number III**). Parallel and antiparallel sheets differ in their hydrogen bond geometries, with bond pairs directly aligned in antiparallel sheets and slightly offset in parallel sheets. Nevertheless, both types of beta-sheets have hydrogen bonds between adjacent backbones.

(Number I) Because *antiparallel* strands run in *opposite* directions, they may be linked by a short sequence of amino acids called a **beta-turn** that induces a 180° bend in the polypeptide chain. Parallel beta-strands do not reverse directionality, so neighboring strands must instead be linked by longer loops that make 360° turns to align the N-terminal regions of neighboring strands. Parallel strands such as those in ataxin-3 can never be linked by beta-turns.

Educational objective:

Parallel and antiparallel beta-sheets differ in the alignment of their strands, with parallel strands having aligned N-termini. Antiparallel strands can be linked by 180° beta-turns whereas parallel strands require longer connecting regions. All beta-sheets participate in backbone hydrogen bonds.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401040

System:
Amino Acids and Proteins